

Flask + PM2 + HTTPS Deployment Guide for oe.snttechverse.com

Generated on: 2025-07-18 05:55:15

1. Overview

This guide provides step-by-step instructions to deploy a Flask application using PM2, with HTTPS enabled via Let's Encrypt, reverse proxied through Nginx, and served through a Cloudflare-managed domain (oe.snttechverse.com).

2. Prerequisites

- A Linux server (Ubuntu recommended)
- Python 3 and Flask app (run.py)
- Domain oe.snttechverse.com added in Cloudflare
- PM2 installed
- Nginx and Certbot installed

3. Flask App Setup with PM2

Edit run.py to run on localhost:

```
app.run(host='127.0.0.1', port=8000)
```

Start with PM2:

```
pm2 start run.py --interpreter python3 --name flaskapp
```

```
pm2 save
```

```
pm2 startup
```

4. Configure Cloudflare DNS

- Go to Cloudflare dashboard > DNS

- Add an A record:

Name: oe

IPv4: Your Server IP

Proxy: 'DNS only' initially (orange cloud off)

Flask App HTTPS Setup with Cloudflare & PM2

5. Install Nginx and Certbot

Install:

```
sudo apt update  
sudo apt install nginx certbot python3-certbot-nginx
```

6. Nginx Config for Flask App

Create config:

```
sudo nano /etc/nginx/sites-available/oe.snttechverse.com
```

Paste:

```
server {  
    listen 80;  
    server_name oe.snttechverse.com;  
    location / {  
        proxy_pass http://127.0.0.1:8000;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
    }  
}
```

Enable and test:

```
sudo ln -s /etc/nginx/sites-available/oe.snttechverse.com /etc/nginx/sites-enabled/  
sudo nginx -t  
sudo systemctl restart nginx
```

7. Issue Let's Encrypt Certificate

Run:

```
sudo certbot --nginx -d oe.snttechverse.com
```

Follow the prompts.

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To auto-renew:

```
sudo crontab -e
```

```
Add: 0 0 * * * /usr/bin/certbot renew --quiet
```

8. Update Cloudflare

- Change A record proxy status to 'Proxied' (orange cloud ON)
- In SSL/TLS settings > Set mode to 'Full'

9. Debugging Common Issues

- Nginx not starting: check with 'sudo nginx -t'
- Port conflict: ensure Flask is not running on port 80 directly
- Certbot errors: ensure correct domain and remove broken SSL config
- Nginx service inactive: start with 'sudo systemctl start nginx'

10. Verify Setup

- Access: <https://oe.snttechverse.com>
- Use SSL Labs or curl to confirm HTTPS response
- Check PM2 status: `pm2 ls`
- View logs: `pm2 logs flaskapp`